

Clinical therapeutics in feline medicine: updates for old and new drugs

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Abstract

Practical relevance:

Over the past 10 years, feline medicine has significantly advanced through the addition of new pharmaceuticals and alternative formulations available for cats, as well as improvements in the knowledge about existing agents. Through continued drug development, the therapeutic options for cats are expanding.

Clinical challenges:

There are many species-specific clinical challenges encountered with the treatment of feline diseases. Additionally, therapeutic options can vary based on geography and change over time.

Evidence base:

This review article discusses the current evidence for some of the newer therapeutic agents that are either presently available for use by feline medicine practitioners or are in development and undergoing clinical trials. Their recent introduction means the evidence for some of these medications is still emerging. The clinical use of these agents, available pharmacokinetic and pharmacodynamic data, and reported adverse effects in cats are presented.

Audience:

The information in this article is relevant to all veterinarians who practice feline medicine.

Introduction

There is an increasing need for medications that can effectively manage both acute and chronic diseases in cats, and recently, there has been an influx of new pharmaceuticals on the market that are available for the treatment of various feline medical conditions. Advancements in clinical pharmacology are also assisting veterinarians with overcoming the unique challenges experienced when diagnosing and treating cats, including patient and client compliance. Consequently, the aim of this article is to review a number of the pharmaceuticals pertinent to feline medicine that are either emerging or have new clinically applicable information available since our previous review.¹ Approved uses for these medications may vary based on geographic location and may also change over time. Therefore, the authors recommend that clinicians are knowledgeable about the drugs available and approved within their geographic area and reference their respective drug licensing authority for additional information.

Buprenorphine

Buprenorphine is a lipophilic opioid derivative that results in mild to moderate sedation or pain control in cats (Table 1), with the potential for fewer systemic adverse effects compared with other opioids.² After systemic absorption of buprenorphine, which is a partial mu receptor agonist and kappa receptor antagonist, it is metabolized in the liver via N-dealkylation by cytochrome P450 3A4 to form norbuprenorphine, an active metabolite. Buprenorphine and norbuprenorphine undergo glucuronidation, forming inactive metabolites, which are excreted mainly in feces via bile and partially in urine.^{2,3} Other feline-specific pharmacologic data are listed in Table 2.

Table 1

Recommended drug dosages, reported adverse effects and clinical indications in cats of the therapeutics reviewed in this article

Drug	Dosage	Adverse effects	Clinical indications* and other comments
Buprenorphine (prolonged duration) ❖ Transdermal 20 mg/ml (Zorbum; Elanco Animal Health) ❖ Subcutaneous high concentration 1.8 mg/ml (Simbadol; Zoetis)	❖ 2.7-6.7 mg/kg applied topically (dorsal cervical area) 1–2 h before surgery ❖ 0.24 mg/kg SC applied ~1 h before surgery	❖ Hyperthermia, mydriasis, euphoria, tachycardia, hyperactivity, dysphoria, sedation, anorexia ❖ Hyperthermia, mydriasis, euphoria, tachycardia, hyperactivity, dysphoria, sedation, anorexia	❖ Postoperative analgesia for 4 days. ^{4,5} Should not be applied to diseased or injured skin. ⁴ ❖ Postoperative analgesia for 24 h. Daily application for 3 days. ⁵
Pregabalin Oral solution 50 mg/ml (Bonqat; Zoetis)	5 mg/kg PO given ~1.5 h before an anticipated stressful event	Ataxia, lethargy, vomiting, proprioception abnormalities, muscle tremors, anorexia, weight loss	Acute anxiety ¹⁵

Drug	Dosage	Adverse effects	Clinical indications* and other comments
Gabapentin	❖ Sedation: 100 mg/cat or 10-30 mg/kg PO ~2 h before an anticipated stressful event ❖ Analgesia: 10 mg/kg PO q12h¹⁶	Sedation, ataxia, behavioral changes, muscle tremors, elevated third eyelid, mydriasis, anisocoria, vomiting/diarrhea, hypersalivation	❖ Sedation, anxiety, as part of multimodal analgesia ❖ Consider dose reduction in cats with CKD
Frunevetmab (Solensia; Zoetis)	1–2.8 mg/kg SC once every 28 days for cats ≥2.5 kg and older than 7 months of age	Dermatitis, pruritus, alopecia	Osteoarthritic analgesia ^{17–19}
Bexagliflozin Oral tablets 15 mg (Bexacat; Elanco Animal Health)	15 mg/cat PO q24h	Euglycemic diabetic ketoacidosis, diarrhea, vomiting, anorexia, urinary tract infections	DM ^{20,21} See Table 3 for contraindications
Velagliflozin Oral solution 15 mg/ml (Senvelgo; Boehringer Ingelheim Animal Health)	1 mg/kg PO q24h	Euglycemic diabetic ketoacidosis, diarrhea, vomiting, anorexia, urinary tract infections	DM ^{20,21} See Table 3 for contraindications
Remdesivir Intravenous solution (Veklury/compounded formulations)	Recommended q24h IV dosages based on the form of FIP or site of infection[†] ❖ Effusive: 15 mg/kg ❖ Non-effusive: 15 mg/kg ❖ Ocular: 15-20 mg/kg ❖ CNS: 20 mg/kg	Anaphylactic reactions, fluid overload	FIP ^{22–26}
GS-441524 Compounded oral formulation	Dosing once or split twice daily according to response and ease of administration[‡] ❖ Effusive: 7.5 mg/kg (minimum recommended dose; adjust dosing according to response up to 15 mg/kg/day) ❖ Non-effusive: 7.5 mg/kg ❖ Ocular: 7.5-10 mg/kg ❖ CNS: 10 mg/kg q12h	Increased liver enzymes (ALT), diarrhea	FIP ^{22–26}
Molidustat Oral suspension 25 mg/ml (Varenzin-CA1; Elanco Animal Health)	5 mg/kg PO q24h for up to 28 days [‡] After 28 days of treatment a pause of ≥7 days is required	Vomiting, ²⁷ systemic hypertension, mild hyperkalemia, thromboembolism	Non-regenerative anemia associated with CKD ²⁷
Telmisartan Oral solution 10 mg/ml (Semintra; Boehringer Animal Health)	❖ Systemic hypertension: 1.5 mg/kg PO q12h for 14 days, then 2 mg/kg PO q24h with dose reductions of 0.5 mg/kg as indicated vs 2 mg/kg PO q24h ❖ Proteinuria: 1 mg/kg PO q24h	GI upset, lethargy, acute kidney injury, hypotension, increased liver enzymes	Systemic hypertension and proteinuria ^{28,29}
Tamsulosin	0.004–0.006 mg/kg PO q12–24h	Hypotension, vomiting, allergic reaction	Ureterospasms ^{30,31}
Rivaroxaban	0.5–1 mg/kg/day or 2.5–5 mg/cat PO q24h	Vomiting, increased risk of spontaneous hemorrhage	ATE ^{32,33}
Capromorelin Oral solution 20 mg/ml (Elura; Elanco Animal Health)	2 mg/kg PO q24h	Vomiting, hypersalivation, inappetence, behavior changes, lethargy, anemia, dehydration	Weight loss associated with CKD ³⁴
Rapamycin Felycin alpha-CA1, sirolimus delayed-release tablets; TriviumVet)	0.3 mg/kg PO once weekly [§]	DM, immunosuppression, cytopenias, GI upset, polydipsia, arrhythmias, urinary incontinence	Ventricular hypertrophy with subclinical HCM ^{35–37}

ATE = aortic thromboembolism; ALT = alanine transaminase; CNS = central nervous system; DM = diabetes mellitus; CKD = chronic kidney disease; FIP = feline infectious peritonitis; GI = gastrointestinal; HCM = hypertrophic cardiomyopathy

*

Clinical indications listed are approved uses or extra-label if there are no uses that are currently approved. See the text for more detail

†

Remdesivir must be diluted in 100 ml of saline and given slowly as an intravenous constant rate infusion over 4 h, q24h. Compounded formulations of remdesivir available legally in some countries are administered as a diluted solution and given over 20 mins. Remdesivir administration is given during hospitalization over the initial 72 h or until the patient can accept oral medications

‡

Compounded GS-441524 is given PO q24h or divided and given q12h for 84 days, with the dosage adjusted according to response. In cats with effusive FIP, shorter treatment durations (42 days) are reported as effective.³⁸ Oral GS-441524 should be administered on an empty stomach, followed by a small amount of water or wet food, with a full meal only given 30–60 mins later

§

Molidustat and rapamycin are currently conditionally approved for use by the US Food and Drug Administration. Similar to fully approved drugs, their safety has been proven when used according to the label; however, a ‘reasonable expectation of effectiveness’ has been shown opposed to ‘substantial evidence’ required for full approval. These drugs may only be prescribed for labeled use pending a full demonstration of effectiveness for full approval

Table 2

Summary of the clinically relevant available pharmacokinetic data in cats for the therapeutics reviewed in this article³⁹

Drug	Route of administration	Bioavailability	Peak plasma concentrations	Half-life
Buprenorphine	Transdermal	16%	7.3–20 h	65–90 h
Buprenorphine	SC (high concentration)	91%	NA	12.3 h
Pregabalin	PO (solution)	94%	20–60 mins	14.7 h
Gabapentin	PO	90%	65–100 mins	2.8–3.8 h
Frunevetmab	SC	60–73%	3.5 days	10–12 days
Bexagliflozin	PO (tablets)	NA	0.5–2 h	4–6.6 h
Velagliflozin	PO (solution)	NA	0.25 h	3.7–6.4 h
Remdesivir*	IV	NA	1 h	5.1 h
GS-441524	PO (compounded)	NA	1.5 h	3.8–8.6 h
Molidustat	PO (suspension)	NA	1 h	5–6 h
Telmisartan	PO (solution)	NA	30 mins	8 h
Rivaroxaban	PO	NA	2.5 h	7.5 h
Capromorelin	PO (solution)	NA	0.25–1 h	1.1 h

NA = not available

*

Remdesivir pharmacokinetic information is based on GS-441524 concentration due to its rapid conversion to GS-441524⁴⁰

Historically, buprenorphine has been administered either parenterally or via the buccal transmucosal route in cats, requiring repeated dosing within 24 h. Recently approved alternative formulations provide prolonged durations of analgesia in cats via a transdermal solution at 20 mg/ml (Zorbiu; Elanco Animal Health) and a high-concentration formulation for subcutaneous injection at 1.8 mg/ml (Simbadol; Zoetis) (Table 1).^{4,5} These alternative routes of administration can improve compliance compared with daily administration of oral medications to cats.

The transdermal buprenorphine solution, formulated with an absorption enhancer, is applied to the dorsal cervical area. It is sequestered in the stratum corneum, allowing for prolonged systemic delivery. The transdermal solution should not be applied to other locations or diseased or injured skin as absorption characteristics may differ.⁴ In an efficacy study, 89/112 cats receiving transdermal buprenorphine following elective neutering and forelimb onychectomy did not require rescue analgesia or opioid reversal or experience an adverse event during the 96 h postoperative period.^{4,6} This proportion of cats was higher than those that were treated with placebo (42/107 cats), although it is important to note that both groups received a standardized treatment of other pre-emptive analgesia (dexmedetomidine hydrochloride and a lidocaine metacarpal four-point ring block).

The subcutaneous injection takes advantage of buprenorphine's slow dissociation from the mu receptor and its ceiling effect to limit adverse effects, despite its delivery at a higher dose (Table 1).⁵ In an efficacy study, 66/93 cats that received high-concentration buprenorphine subcutaneously did not require rescue analgesia compared with 45/102 cats that received placebo in the 72 h postoperative period after soft tissue surgery (all cats received acepromazine with or without glycopyrrolate as premedication).^{5,7} High-concentration buprenorphine also resulted in similar analgesic effects in cats after dental extractions compared with buprenorphine hydrochloride 0.3 mg/ml given at a dosage of 0.02 mg/kg IM q8h (all cats also received meloxicam and dental nerve blocks with bupivacaine).⁸

The frequency of administration of buprenorphine, which is appropriate for treating mild to moderate pain, should be based on the expected duration of the noxious stimulus.⁹ In the case of surgery, pain can be prolonged postoperatively (2–3 days or longer), and this duration should be considered by clinicians given the importance of preventive analgesia.⁹ Subcutaneous and transdermal delivery of buprenorphine offer ease of administration to prolong the duration of analgesia; however, compared with intravenous and intramuscular administration, these methods might be inferior for postoperative analgesia.¹⁰ The bioavailability of subcutaneous and transdermal buprenorphine is comparatively low and highly variable among individuals.^{11,12} Due to the variable absorption of these newer formulations, as well as anecdotal clinical experience suggesting prolonged adverse effects in cats treated at labeled dosages, more conservative (or lower) dosing may be needed in some cats.

The adverse effects associated with the administration of the new formulations of buprenorphine are summarized in Table 1.^{4,5} Sedation and transient euphoria are the most common behavior changes associated with buprenorphine administration in cats.¹³ Compared with other species, cats also exhibit unique responses to opioids, including mydriasis and hyperthermia.¹³ While in one study cats that received transdermal buprenorphine at the higher end of the dosing range exhibited no increase in adverse effects but a more profound analgesic effect compared with placebo,⁶ in another study evaluating the safety of the transdermal buprenorphine solution, hyperthermia was observed in 66% of the cats on the day after surgery.¹⁴ Clinically, these adverse effects may limit the use of buprenorphine in some cats, and repeated naloxone dosing may be needed to reverse them due to the combination of naloxone's short half-life and the prolonged duration of effect of the newer formulations. Safe use of these longer-duration buprenorphine formulations has not been assessed in cats with renal, hepatic, cardiac or respiratory disease or in cats that are pregnant, lactating, intended for breeding or younger than 4 months of age.^{4,5}

Pregabalin

Pregabalin is primarily indicated for reducing acute anxiety in cats. It is structurally related to the inhibitory neurotransmitter gammaaminobutyric acid (GABA), but it does not interact with or bind to the GABA receptor. Pregabalin is a ligand of the presynaptic alpha-2-delta (α -2- δ) subunit of voltage-gated calcium channels within the central nervous system (Figure 1).

Figure 1

A schematic representation of the mechanism of action of pregabalin and gabapentin. Pregabalin and gabapentin are ligands of the presynaptic alpha-2-delta (α -2- δ) subunit of voltage-gated calcium (Ca^{2+}) channels within the central nervous system. They reduce presynaptic neuronal calcium influx, thus decreasing the subsequent release of many excitatory neurotransmitters, including glutamate, norepinephrine, substance P and calcitonin gene-related peptide. These mechanisms account for pregabalin and gabapentin's anxiolytic, analgesic and antiepileptic properties.^{46,47}

A pregabalin oral solution (Bonqat; Zoetis) is now available and labeled for cats to minimize fear and anxiety associated with veterinary visits (Table 1). Ease of administration is improved through the small volume required and flavoring of the liquid suspension. Extra-label uses of pregabalin include the treatment of neuropathic pain, as an adjunct analgesic therapy for allodynia and hyperalgesia, and as a substitute for gabapentin as an adjunct perioperative sedation agent.^{41,45}

Pregabalin oral solution is labeled as a single oral dose administered approximately 1–2 h before an anticipated stressful event.¹⁵ In both a field study and a randomized, placebo-controlled study, approximately half of the cats that received pregabalin 1.5 h before a veterinary visit had a good to excellent response to the stress of transport and handling, with decreased vocalization, restlessness and panting.^{15,46}

Pregabalin solution has high oral bioavailability (Table 2)^{43,47} and it can be given with a small amount of food. Renal elimination of the active drug and its metabolites in other species suggests caution in cats with kidney disease. It should also be used cautiously in cats with cardiac disease or hypertension as it may cause bradycardia and reflex hyper-tension.¹⁵ The most common adverse effects associated with pregabalin oral solution in cats are listed in Table 1.¹⁵

Gabapentin

Gabapentin is an analog of GABA and is used for its sedative and anxiolytic effects and for the treatment of musculoskeletal pain in cats (Table 1).^{48,54} Despite its frequent and longterm use, the mechanism of action of gabapentin is not completely understood.⁵⁵ Gabapentin does not interact with GABA type-B receptors and, until recently, was not known to influence GABA type-A

receptors.^{56,57} Similarly to pregabalin, it binds to the α -2- δ subunit of voltage-gated calcium channels (Figure 1), and its inhibitory effects on voltage-gated calcium influx are believed to be responsible for many of its therapeutic applications.^{55,58}

Oral gabapentin is highly bioavailable when administered on an empty stomach (Table 2).^{48,59} Repeated dosing (10 mg/kg PO q12h for 14 days) does not alter its pharmacokinetics but is ineffective at maintaining the plasma concentrations correlated with efficacy in other species,⁴⁸ although the optimal plasma concentrations of gabapentin for clinical efficacy in cats are not yet fully understood. In one study, transdermal gabapentin (Lipoderm base) was found to be systemically absorbed in cats but reached lower plasma concentrations than oral administration.⁶⁰ An improvement in pain scores was also observed; however, the fact that the assessment was not masked should be considered. Gabapentin plasma concentrations may increase over several days with transdermal administration, but the time to reach a steady state with this route is unknown due to the short time frames that have been studied.^{48,60} Cats with chronic kidney disease (CKD) have higher dose-normalized serum gabapentin concentrations compared with healthy cats, providing evidence that dose reduction is needed for cats with reduced kidney function; however, the optimal dose reduction strategy is not yet established for these patients.⁶¹

The pharmacodynamic effects of gabapentin have been evaluated in multiple clinical trials in cats. Although there are currently no approved indications for gabapentin in cats, it is frequently prescribed, with its main clinical uses being analgesia, as an anxiolytic to reduce stress and improve compliance during veterinary evaluations, and to decrease the minimum alveolar concentration of gas anesthetics.^{50,52,62,65} Gabapentin may also stimulate appetite; one study showed that at a dose of 5 mg/kg, it improved appetite similarly to mirtazapine in cats immediately post-ovariectomy.⁶²

Studies evaluating gabapentin's effectiveness in anesthesia premedication protocols are limited. When administered alongside buprenorphine, gabapentin provided similar levels of preanesthetic sedation to pregabalin and buprenorphine.⁴⁴ Gabapentin also resulted in similar sedation and stress scores as alprazolam in healthy cats prior to ovariohysterectomy.⁶⁶ In further studies, while oral gabapentin administered to cats prior to general anesthesia was found to decrease the minimum alveolar concentration of isoflurane, intravenous gabapentin was not shown to produce the same effect; the reason for this discrepancy remains unclear.^{67,68}

The evidence for gabapentin's effectiveness as an analgesic is weak,^{50,53,62,65} and stronger evidence is available to support its use as a sedative and anxiolytic. It reduces fear responses in cats in trap–neuter–return programs and accelerates behavior modification.^{49,52} Multiple studies have shown that a single dose of gabapentin, administered approximately 2 h prior to a veterinary visit, reduces stress and improves compliance.^{51,54,69,71} If further management of stress is needed after 2 h, clinicians could consider either an additional dose of gabapentin or other therapies depending on the original dose given (an oral dose of gabapentin of up to 47.6 mg/kg has been reported⁷¹), patient compliance and the availability of time. Sedation does not appear to increase when gabapentin given orally at 10 mg/kg is combined with trazadone given orally at 5 mg/kg compared with trazadone alone.⁷² However, another study found that sedation was increased when these drugs were combined and administered at approximately twice the aforementioned doses.⁷³ Gabapentin can result in a mild decrease in serum cortisol concentration compared with placebo,^{74,75} but another study did not corroborate this finding.⁵¹ Serum glucose concentrations do not appear to decrease with gabapentin.^{51,74} The evidence for administering gabapentin to cats supports its use as a sedative but not for analgesia. If prescribed as an analgesic, it should be used as a part of multimodal therapy and not as a sole agent.

Although gabapentin is useful for improving patient compliance and reducing stress during veterinary evaluations, clinicians should be aware that it may influence some clinical parameters. In a randomized, masked clinical trial, gabapentin (100 mg/cat) was shown to impact neurologic examination findings in healthy cats.⁷⁶ Gabapentin did not alter the level of consciousness, cranial nerve function or spinal reflexes, but did cause proprioceptive ataxia and abnormal postural reactions. It has also been found to not affect horizontal pupil diameter, intraocular pressure, Schirmer tear test and intradermal allergy testing results.^{51,77} Gabapentin did not appear to affect systolic blood pressure in a small study of five healthy cats, but a median decrease of 12 mmHg was observed in both healthy cats and those with CKD in another study.^{78,79} Gabapentin is considered appropriate for sedation prior to cardiac evaluation, since studies investigating its effects on echocardiographic parameters have shown either no changes or changes that were not thought to be clinically meaningful.^{71,80,81} Aside from its intended sedative effects, gabapentin's additional reported adverse effects are summarized in Table 1.^{48,54,69,76,81}

Frunevetmab

Frunevetmab (Solensia; Zoetis) is indicated for the management of osteoarthritic pain in cats (Table 1). It is a feline-specific monoclonal antibody against nerve growth factor (NGF; see 'Role of nerve growth factor in osteoarthritic pain' box) that binds to NGF, blocking NGF-mediated pain transmission (Figure 2). Frunevetmab is given as a subcutaneous injection every 28 days, which is an attractive route and frequency of administration for many patients.

Figure 2

A simplified schematic representation of the role of nerve growth factor (NGF) in osteoarthritic pain. NGF, which functions as a signaling protein and is the target of NGF inhibitors such as frunevetmab, binds the high-affinity NGF-specific tropomyosin receptor kinase A (TrkA) receptor of sensory neurons. The NGF–TrkA complex is endocytosed and transported to the neuronal cell body within the dorsal root ganglion where it increases the expression of cell surface receptors (ie, transient receptor potential vanilloid 1 [TRPV1], acid-sensing ion channels [ASIC], bradykinin receptors [BR]), ion channels (ie, voltage-gated sodium channels [VGSC]) and transcription factors (ie, substance P [SP], calcitonin gene-related peptide [CGRP]) central to nociception.⁸⁷ The NGF inhibitor frunevetmab is a feline-specific IgG monoclonal antibody that binds NGF, making it unavailable to bind the TrkA receptor of sensory neurons, thereby stopping the transmission of noxious stimuli mediated by the overproduction of NGF associated with osteoarthritic pain.

Frunevetmab has a half-life of 10–12 days in cats (Table 2). To reach steady-state trough frunevetmab concentrations, two doses administered 28 days apart are required, although clinical responses have been reported within 3–4 weeks.^{17,18,82} In initial studies evaluating frunevetmab in client-owned cats with osteoarthritis, a single dose of frunevetmab improved activity and pain scores for 6 weeks following treatment compared with placebo, without adverse effects.⁸² In multi-dose studies in cats with osteoarthritis treated with frunevetmab or placebo, owners reported a clinical improvement in activity, there was increased activity measured by accelerometer and the veterinarian-assessed joint pain scores were lower in the frunevetmab-treated cats compared with placebo.^{18,19} Studies evaluating the efficacy of frunevetmab compared with non-steroidal anti-inflammatory drugs (NSAIDs) have not been reported to the authors' knowledge.

In people, concerns have been raised about the development of rapidly progressive osteoarthritis in individuals concurrently treated with NSAIDs and monoclonal antibodies against NGF.⁸³ The safety of frunevetmab combined with NSAIDs has not been evaluated. Recently, a case of rapidly progressive osteoarthritis was reported in a dog treated with bedinvetmab, a monoclonal antibody against NGF approved for canine use;⁸⁴ however, this diagnosis has been challenged.⁸⁵ Furthermore, a recent disproportionality analysis evaluating musculoskeletal adverse events in dogs treated with beninvetmab revealed that ligament/tendon injuries, polyarthritis, fractures, musculoskeletal neoplasia and septic arthritis were reported nine times more frequently in bedinvetmab-treated dogs compared with the dogs treated with six comparator drugs (all NSAIDs with the same indication) combined.⁹¹ Additionally, an expert panel concluded a strong suspicion of an association between bedinvetmab and accelerated joint destruction in 19 cases.⁹¹ These reports in people and dogs should alert feline medicine practitioners to the potential of musculoskeletal adverse events secondary to frunevetmab.

The most common adverse effects reported in cats treated with frunevetmab are dermatologic (Table 1).^{18,19} In a recent case series, five cats developed pruritus not associated with the injection site 3–18 days after receiving frunevetmab;⁹² interestingly, four of the five experienced the skin reaction after the first dose of frunevetmab. The initial pruritus and subsequent self-trauma described in these cases may suggest that some cats treated with frunevetmab experience adverse cutaneous sensations like those reported in people treated with NGF inhibitors.^{92,95}

With limited clinical use and studies reporting treatment durations of ≤ 3 months, the long-term impact of frunevetmab on sensory and sympathetic nerve development is unknown. In non-human primates, prolonged exposure to high doses of NGF inhibitors reduced the size of post-ganglionic neuronal cell bodies, which returned to normal after discontinuation of the NGF inhibitor, suggesting these observed changes were reversible.⁹⁶ Frunevetmab appears well tolerated as an alternative therapy for cats with osteoarthritis, although some cats may experience skin reactions after the first injection. Additional research is needed to provide guidance on the long-term use of frunevetmab in cats, especially those with underlying CKD. Close monitoring during treatment is recommended to evaluate for potential adverse effects and each cat's clinical response, which might be facilitated by incorporating the use of pain scoring and owner questionnaires. No information regarding frunevetmab use in cats with a history of immune-mediated disease is available to guide its use in these patients.

Bexagliflozin/velagliflozin

In people with type 2 diabetes, cardiac or renal disease, sodium-glucose-linked transporter 2 (SGLT2) inhibitors have become widely used, which has sparked an interest in the use of these in veterinary medicine. This interest is particularly high in cats due to the similarities between feline diabetes mellitus and type 2 diabetes in people.⁹⁷ SGLT2 inhibitors are a potential oral option for treating diabetes mellitus in cats as they block glucose reabsorption from the glomerular filtrate and increase urinary glucose excretion, thereby lowering blood glucose concentrations (Figure 3). SGLT2 inhibitors also block sodium reabsorption in the renal proximal tubules and play a beneficial role in treating non-diabetic cardiac and renal diseases. In people, SGLT2 inhibition-induced glucosuria decreases plasma uric acid levels, reducing the risk of cardiovascular complications and congestive heart failure.¹⁰³ In addition, SGLT2 inhibition-induced natriuresis decreases plasma volume and blood pressure, in turn reducing cardiac preload and afterload, respectively.¹⁰³ SGLT2 inhibition-induced natriuresis in people promotes sodium delivery to the macula densa, activating tubuloglomerular feedback, afferent vasoconstriction, reduced glomerular pressure and decreased albuminuria, although healthy cats experience a paradoxical induction of glomerular hyperfiltration with dapagliflozin.^{103,104}

Figure 3

A schematic representation of the functional unit of the kidney, the nephron, is depicted with a focus on the sodium-glucose-linked transporters (SGLTs; SGLT1 and SGLT2) along the nephron and the mechanism of the SGLT2 inhibitors. Most of the filtered glucose is reabsorbed by the kidneys within the proximal renal tubules via two types of SGLTs – SGLT-1 and SGLT-2 – that cotransport glucose and sodium into the tubular epithelial cells.⁹⁸ Glucose is then transported across the basolateral membrane by glucose transporter 2 (GLUT2).⁹⁸ The SGLTs cotransport sodium via a carrier protein down the concentration gradient established by the Na-K ATPase pump and actively transport glucose. SGLT2 is a low-affinity, high-capacity glucose transporter located in the early proximal renal tubule epithelium, responsible for 90% of glucose uptake from the glomerular filtrate, whereas SGLT1 is a high-affinity, low-capacity glucose transporter located later in the renal proximal tubule, reabsorbing 10% of the filtered glucose.^{98,99} SGLT1 plays a more significant role in dietary glucose absorption via its presence on the luminal surface of the small intestinal epithelium. SGLT2 inhibitors decrease renal sodium and glucose reabsorption, thereby inducing glucosuria, natriuresis and osmotic diuresis.^{20,21,100,101} The SGLT2 inhibitors available for cats include bexagliflozin and velagliflozin

Recently, two veterinary-approved SGLT2 inhibitors have become available for use in cats: bexagliflozin (Bexacat; Elanco Animal Health) and velagliflozin (Senvelgo; Boehringer Ingelheim Animal Health) (Tables 1–3). The availability of an SGLT2 inhibitor in both tablet form and as a liquid suspension provide options for oral administration to cats. Both agents are highly selective SGLT2 inhibitors, and their glucosuria-induced effects rapidly decrease blood glucose concentrations independently of endogenous or exogenous insulin, thereby promoting a euglycemic state.^{97,103} Bexagliflozin and velagliflozin may also improve insulin sensitivity and support beta (β)-cell recovery.^{97,105} Although bexagliflozin and velagliflozin are highly selective SGLT2 inhibitors, they may partially inhibit SGLT1, which is responsible for sodium and glucose absorption at the small intestinal lumen and late renal proximal tubular epithelium. This inadvertent SGLT1 inhibition may contribute to gastrointestinal adverse effects.^{100,101}

Although SGLT2 inhibitors decrease blood glucose concentrations, clinical hypoglycemia is uncommon.^{106,107} However, cats remain at risk for diabetic ketoacidosis (DKA), which can manifest as euglycemic DKA, especially when the underlying cause for insulin resistance persists. In such cases, a relative lack of insulin continues to drive lipolysis and ketogenesis. Additionally, SGLT2 is expressed on α-cells of the pancreas, which contributes to the risk of developing DKA as SGLT2 inhibitors promote glucagon release.⁹⁷ Ideally, cats treated with SGLT2 inhibitors must have some β-cell function for endogenous insulin production to reduce the risk of developing DKA.^{97,106,107}

It is important to carefully screen cats for candidacy of treatment with an SGLT2 inhibitor (Table 3). Cats with a new diagnosis of diabetes mellitus that are apparently healthy and without major comorbidities are currently considered the most appropriate candidates. Specific contraindications vary with the drug labels,^{20,21} but in general an SGLT2 inhibitor should not be used in a cat with ketosis, pancreatitis or other disease that can increase the risk of developing DKA. The safety and efficacy of SGLT2 inhibitors with CKD, especially advanced-stage CKD (International Renal Interest Society stages 3 and 4), is unknown due to concerns about dehydration and decreased glucose excretion from nephron loss leading to decreased effectiveness.⁹⁷ During treatment, close clinical monitoring is necessary, as with all diabetic cats, to optimize treatment and to minimize adverse effects, including the development of euglycemic DKA (see ‘Monitoring cats treated with bexagliflozin/velagliflozin’ box).

Table 3

Comparison of bexagliflozin and velagliflozin used for glycemic control in cats with diabetes mellitus that are not on or have not been previously treated with insulin therapy^{20,21,100,102}

SGLT2 inhibitor	Formulation	Pharmacokinetics	Contraindications
Bexagliflozin (Bexacat; Elanco Animal Health)	Flavored tablets	❖ Food increases oral absorption ❖ Metabolism extrapolated from people, including glucuronidation and oxidation ❖ Unchanged drug excreted in feces ❖ Glucuronide metabolite excreted in urine	❖ Cats previously or currently treated with insulin ❖ Insulin-dependent diabetes mellitus ❖ DKA ❖ Anorexia, dehydration or lethargy ❖ Clinical pancreatitis ❖ Hepatic disease (bexagliflozin) or bilirubin >0.5 mg/dl (velagliflozin) ❖ Reduced renal function (bexagliflozin) or creatinine >2 mg/dl (velagliflozin) ❖ Chronic or unresponsive diarrhea (velagliflozin) ❖ Cachexia (velagliflozin)
Velagliflozin (Senvelgo; Boehringer Ingelheim Health)	Oral solution	❖ Food decreases oral absorption ❖ Metabolism extrapolated from people, including glucuronidation and oxidation ❖ Unchanged drug excreted in feces ❖ Glucuronide metabolite excreted in urine	

DKA = diabetic ketoacidosis; SGLT2 = sodium-glucose-linked transporter 2

From available clinical studies in otherwise healthy, newly diagnosed diabetic cats that were not previously treated with insulin, velagliflozin and bexagliflozin lowered blood glucose concentrations in most cats.^{106,107} However, up to 50% of cats experienced at least one drug adverse effect, with 5–7% of cats developing DKA. The most common adverse effects of the SGLT2 inhibitors are listed in Table 1.^{106,107} Other studies have begun investigating the extra-label use of bexagliflozin or velagliflozin in poorly controlled diabetic cats already receiving insulin. These studies have reported some success in lowering blood glucose concentrations, reducing exogenous insulin requirements and improving clinical outcomes in many cats.^{99,108,111} In one, a larger study evaluating velagliflozin in historically poorly controlled diabetic cats receiving insulin, treatment failure or drug adverse effects limited velagliflozin's use in 25% of cats and 7% of cats developed DKA.¹⁰⁸

Remdesivir/GS-441524

Remdesivir is a broad-spectrum antiviral effective against RNA viruses, including feline coronavirus (FCoV). It is a prodrug of the nucleoside analog, GS-441524, which is further phosphorylated into its active nucleoside triphosphate (NTP) metabolite that then interferes with coronavirus replication (Figure 4).^{22,112} Remdesivir and GS-441524 have shown efficacy in treating FCoV associated with feline infectious peritonitis (FIP) based on in vitro studies, animal models and the treatment of FIP in cats.^{23,113,117} Antiviral protocols, including parental remdesivir used in combination with oral compounded GS-441524 or oral GS-441524 alone, vary depending on the cat's clinical signs, availability of reputable drug formulations and ease of drug administration to the patient. Recommended dosages for remdesivir and oral GS-441524 also depend on the cat's clinical signs and are summarized in Table 1.²⁴ The use of remdesivir and oral compounded GS-441524 allows veterinarians to develop more standardized protocols to treat cats with FIP successfully and alleviates the need for owners to rely on unlawful GS-441524.^{24,114}

Figure 4

A schematic representation of the relationship between the antivirals, remdesivir and GS-441524, and their mechanism of action. (a) Remdesivir is a prodrug of the nucleoside analog that aids in attaining intracellular concentrations of GS-441524. GS-441524 is further phosphorylated to its active nucleoside triphosphate (NTP) metabolite that acts as a nucleoside analog and inhibits the RNA-dependent RNA polymerase (RdRp) of coronaviruses.^{22,112} (b) Chemical structure comparison of the active metabolite of remdesivir and GS-441524, NTP metabolite, to adenosine triphosphate (ATP). (c) Schematic representation of the NTP metabolite competing with ATP for incorporation into replicating viral RNA via RNA-dependent RNA polymerase (RdRP), resulting in the termination of viral RNA synthesis. Coronavirus survival requires functional RNA-dependent RNA polymerase for viral replication. It is the NTP metabolite that acts as an inhibitor of RdRP, interfering with coronavirus replication.^{22,112}

Reports indicate that 75–100% of cats treated with antivirals – remdesivir or GS-441524 – show some clinical improvement within 48–72 h.^{23,25} The clinical response to these antivirals is incredible considering FIP was, until a few years ago, considered an almost uniformly fatal disease for cats.¹¹⁸ Both antivirals appear to be well tolerated in cats over the 84 days of treatment in these studies. Adverse effects reported with remdesivir treatment in cats are listed in Table 1;^{23,25} additionally, the formation of urolithiasis has been reported in cats treated with unlawful GS-441524.^{119,120}

Studies are beginning to emerge that support shorter treatment durations in some cat populations with FIP treated with licensed GS-441524. In a small prospective, randomized controlled study in cats diagnosed with effusive FIP, 42 days of treatment with licensed GS-441524 was effective in resolving clinical signs without noted adverse effects.³⁸ In addition, the use of acute-phase protein – alpha-1 acid glycoprotein or serum amyloid A – monitoring may offer an objective marker of clinical response in cats with FIP. For example, alpha-1 acid glycoprotein offers a non-specific monitoring tool to help clinicians assess FIP recovery and tailor antiviral treatment durations.¹²¹ The use of GS-441524 therapeutic drug monitoring has also begun to be evaluated in cats but, in the absence of feline population pharmacokinetic data, it is not routinely available for clinical decision-making.⁴⁰ Cats with more severe FIP or younger cats may require GS-441524 dosage increases with increasing body weight.

With new antivirals becoming available, the opportunity to successfully treat cats with FIP is promising and continues to evolve. Overall, remdesivir and GS-441524 are well tolerated in most cats. Effective antiviral protocols, including more oral options, will likely continue to develop. Therapeutic options for cats may vary based on clinical presentation, severity of illness and the antivirals available to veterinarians and owners.

Molidustat

Molidustat is indicated for use in cats with non-regenerative anemia associated with CKD and acts by increasing the levels of endogenous erythropoietin, an important hematopoietic growth factor. Erythropoietin is primarily produced by renal peritubular interstitial cells, and its production is stimulated by hypoxia-inducible factor (HIF). During normoxia, HIF is degraded by hypoxia-inducible factor prolyl hydroxylase (HIF-PH) enzymes; when there is renal hypoxia, however, this inhibits HIF-PH enzymes, stimulating erythropoiesis (Figure 5). With CKD comes a decrease in the number of erythropoietin-producing renal peritubular interstitial cells; this leads to the decrease in erythropoietin levels well recognized in cats with this condition. However, a relatively higher oxygen tension in the kidneys remains due to the reduced metabolic activity secondary to CKD. Experimentally, a hypoxic stimulus that inhibits HIF-PH enzymes still results in erythropoiesis,^{122,123} and it is this mechanism that therefore provides the basis for the use of molidustat, an HIF-PH inhibitor, to treat anemia in CKD.

Figure 5

A simplified schematic diagram of the mechanism of action of molidustat. During renal normoxia, hypoxia-inducible factor (HIF) is degraded by hypoxia-inducible factor prolyl hydroxylase (HIF-PH) enzymes that are oxygen sensing. During renal hypoxia, or in the presence of molidustat, these enzymes are inhibited, leading to stabilization of HIF and erythropoiesis. The HIF complex binds to hypoxic response elements of target genes, ultimately leading to endogenous erythropoietin production¹²⁴

Molidustat oral suspension (Varenzin-CA1; Elanco Animal Health) can be administered daily for up to 28 consecutive days based on its conditional approval status (Table 1).²⁷ Its formulation as a liquid suspension improves ease of administration for many cats. Molidustat effectively increased erythropoietin concentrations, as detected by a human enzyme-linked immunosorbent assay, as well as hematocrit in healthy cats.¹²⁵ Minimal drug accumulation occurs, and the reported elimination half-life is 4–6 h, supporting once daily dosing for cats with decreased renal function. In a randomized, masked, placebo-controlled clinical trial, on day 28 of treatment, approximately half of the cats with CKD and concurrent anemia had an increase in hematocrit by either >25% from baseline or an absolute increase of at least 4% when treated with molidustat.¹²⁶

Concurrent iron supplementation with the administration of erythrocyte-stimulating agents is typically recommended in cats. However, most human CKD patients experience a decrease in hepcidin when treated with HIF-PH inhibitors, which may increase the bioavailability of iron.¹²⁴ Young, healthy cats did not develop apparent iron deficiency with molidustat when assessed by cell hemoglobin in reticulocytes.¹²⁶ Iron status has not been assessed in a molidustat-treated feline CKD population, however, and consequently it is currently unknown if cats with anemia associated with CKD treated with HIF-PH inhibitors would benefit from concurrent iron supplementation. Molidustat appears to be relatively well tolerated in both healthy cats and cats with CKD, aside from vomiting.^{125,126} Reported adverse effects of molidustat in cats are provided in Table 1.^{27,126}

It is unknown if HIF-PH inhibitors are more effective than erythrocyte-stimulating agents, such as darbepoetin, for the treatment of anemia in CKD in cats; however, no other therapies are currently approved for treatment in this species. Clinicians should consider multiple factors when deciding on treatment options for anemia in CKD (Table 4). Additionally, recommendations on when to initiate treatment are evolving. Therapy was traditionally recommended when anemia was severe (packed cell volume [PCV] <20%) or if it was affecting the patient's quality of life.^{128,129} However, anemia is a negative prognostic indicator in cats with CKD,^{130,131} and a PCV <27% was recently found to be associated with poorer quality of life scores.¹³² Consequently, clinicians should consider starting treatment of anemia in CKD earlier, with a PCV <28% suggested as a new threshold for cats.¹³³

Table 4

Clinical differences between molidustat and darbepoetin used in cats for treatment of non-regenerative anemia associated with chronic kidney disease

	Molidustat ^{27,125,126}	Darbepoetin ¹²⁷
Mechanism of action	Hypoxia-inducible factor prolyl hydroxylase inhibitor	Erythrocyte-stimulating agent
Approval	Conditionally approved	Extra-label
Route of administration	Oral solution	Subcutaneous injection
Frequency of administration	Daily	Weekly
Adverse effects	❖ Vomiting ❖ Systemic hypertension ❖ Mild hyperkalemia ❖ Thromboembolism	❖ Vomiting ❖ Systemic hypertension ❖ Seizures ❖ Pure red cell aplasia
Cost*	\$	\$\$\$
Iron supplementation	Potentially not required	Recommended

★

Each \$ represents 100 US dollars and is approximated from the cost of a 27 ml bottle of Varenzin-CA1 (Elanco Animal Health; molidustat) and a 1 ml vial of Aranesp (Amgen; darbepoetin alfa) 25 µg/ml from the authors' institutions

Telmisartan

Telmisartan (Semintra; Boehringer Ingelheim Animal Health) is an angiotensin receptor blocker (ARB) used for the treatment of systemic hypertension in cats, as well as for the treatment of proteinuria associated with CKD. It is formulated for cats as a liquid suspension that is rapidly absorbed and can be given fasted or with food.^{28,134} It is a selective ARB due to its affinity for binding to the angiotensin II type 1 receptor and not the angiotensin II type 2 receptor, allowing it to mediate some of the pathologic effects of angiotensin II while sparing certain protective effects (Figure 6).^{135,136} Telmisartan undergoes effective glucuronidation in cats, as indicated by an in vitro study.¹³⁷

Figure 6

A simplified flow diagram of the renin–angiotensin–aldosterone system with effects of stimulation of the classic angiotensin receptor types 1 and 2 by angiotensin II. The angiotensin receptor blocker telmisartan is a selective antagonist of the angiotensin II type 1 receptor. ACE = angiotensin converting enzyme; ARB = angiotensin receptor blocker; AT = angiotensin II type

Telmisartan dosage recommendations for treating systemic hypertension in cats vary between studies (Table 1).^{29,138,140} A dose-finding study in healthy cats and a prospective, randomized, double-masked, placebo-controlled clinical trial evaluating telmisartan therapy for cats with systemic hypertension used an escalating dosage strategy, initially 1.5 mg/kg q24h for 14 days and then 2 mg/kg q24h, with further dose reductions of 0.5 mg/kg for individual patients.^{138,139} A similar study was conducted concurrently that investigated telmisartan at a dosage of 2 mg/kg PO q24h.¹⁴⁰ The cohorts in both studies included cats with systolic arterial blood pressures between 160 mmHg and 200 mmHg. Although dosing strategies differed slightly between these two studies, there was an overall reduction in blood pressure by approximately 21 mmHg by day 14 and approximately 24 mmHg by day 28 of treatment. Additionally, approximately 52% of these cats achieved a reduction in blood pressure to <150 mmHg or a decrease of at least 15% from baseline by day 28. Although evidence is limited for telmisartan's efficacy in cats with a blood pressure of >200 mmHg, it may still be effective.¹⁴¹ For refractory cases, combination therapy with amlodipine may be considered with careful monitoring for hypotension.

Telmisartan was determined to be noninferior to benazepril for the treatment of proteinuria associated with CKD.¹⁴² It significantly reduced the urine protein:creatinine ratio at all timepoints, whereas benazepril did not. Furthermore, a higher percentage of telmisartan-treated cats were reclassified from proteinuric to non-proteinuric or borderline proteinuric (57.2%) compared with the benazepril-treated cats (30.8%) by the end of the study. Telmisartan may be superior to angiotensin-converting enzyme inhibitors for the treatment of proteinuria associated with CKD in cats; however, further studies are needed to verify this and to evaluate its impact on survival.

Reported adverse effects of telmisartan are summarized in Table 1.^{28,138,140,142} In cats receiving telmisartan, if a cat experiences a decreased appetite or anorexia, telmisartan should be discontinued to prevent acute kidney injury.¹³⁹ Although telmisartan appears to be well tolerated by most cats, close monitoring within 1–2 weeks of starting telmisartan or dose escalation for deterioration in renal function and hyperkalemia is recommended, particularly in cats with advanced stages of CKD.

Tamsulosin

Tamsulosin is a selective α_1 -adrenergic antagonist that is used for relaxation of smooth muscle of the urinary tract. It is considered uroselective due to its specificity for α_{1A} - and α_{1D} -adrenergic receptors, which predominate in the urinary tract smooth muscle, compared with α_{1B} -adrenergic receptors, which are found in the vasculature.¹⁴³ As a result, its use should relax smooth muscle in the urinary tract without causing significant systemic hypotension. Currently, there is limited to no data on its pharmacokinetics or pharmacodynamics in cats. Due to the available strength (0.4 mg capsules), compounding is required to achieve the recommended dosing in cats (Table 1). Consequently, it is not uncommon for cats to receive a higher dose in clinical practice.

Smooth muscle relaxation is often used as part of the medical management for feline urethral and ureteral obstructions. Nevertheless, prazosin, another non-selective α_1 -adrenergic antagonist, is currently not recommended¹⁴⁴ based on several recent studies showing no benefit of its use.^{145, 147} There is a paucity of data on tamsulosin in cats. Tamsulosin dosed at 0.1 mg/cat (~0.017 mg/kg) PO q24h for 10 days did not significantly alter urethral pressure profile parameters in a small population of healthy male cats.¹⁴⁸ Its effectiveness in treating urethral obstructions also remains uncertain, especially as obstructions are more common distal to the smooth muscle, in the skeletal muscle of the prostatic urethra.^{149, 151} In contrast to the urethra, the feline ureter contains smooth muscle that surrounds the mucosa throughout its length.¹⁵¹ In a retrospective study of cats with obstructive ureterolithiasis, which can cause acute kidney injury, stone passage was documented in 22/70 (31%) cats treated with tamsulosin as part of their regimen.³⁰ Due to the lack of a comparator in this study, however, the benefit of tamsulosin for medical management of obstructive ureterolithiasis in cats cannot be ascertained, and further research is required. It is also difficult to compare these findings with other studies evaluating the success of medical management for benign ureteral obstructions. In one study, serial imaging was only reported in 11/52 cats with obstructive ureteral calculi that were treated medically;¹⁵² passage of stones was reported in nine of those cases, but an improvement in creatinine was only seen in 4/9 cats. In a more recent study, successful medical management was reported in 17/75 (23%) ureters with obstructive calculi.¹⁵³ No adverse effects of tamsulosin, such as systemic hypertension, vomiting or allergic reaction, were reported in the abovementioned study.³⁰

Rivaroxaban

Rivaroxaban is a direct Xa inhibitor used in cats at risk for thrombosis. Clotting factor X is essential for secondary hemostasis and, when activated (Xa), it converts prothrombin to thrombin (Figure 7). Rivaroxaban directly inhibits Xa, reducing thrombin formation and indirectly inhibiting thrombin-induced platelet aggregation.¹⁵⁴ Having gained a better understanding of the role of thrombin and platelets in arterial and venous thrombosis, rivaroxaban is now used in people for the prevention of both conditions.¹⁵⁴ Rivaroxaban offers advantages such as high oral bioavailability, rapid onset of action and predictable anticoagulant activity, reducing the need for frequent clinical monitoring in people.¹⁵⁵ Similar efficacy and safety of rivaroxaban are anticipated in other species, leading to its extra-label use as an antithrombotic agent for cats (Table 1).^{32, 156, 157}

Figure 7

A limited schematic representation of the coagulation cascade that highlights the key coagulation factors to illustrate the mechanism of action of rivaroxaban. The intrinsic and extrinsic coagulation pathways activate factor X, which is essential for thrombin formation (factor IIa) and clot development through the common coagulation pathway. The critical activated factors of the intrinsic pathway triggered by endothelial damage include XIIa, XIa, IXa and VIIIa. Factor VII of the extrinsic pathway is activated by tissue factor (III). Prothrombin is activated by both factors Xa and Va within the common pathway to produce the thrombin needed to activate fibrinogen into fibrin and, with the help of factor XIIIa, the development of a cross-linked fibrin clot. Rivaroxaban is a direct factor Xa inhibitor that does not require a cofactor to bind Xa to produce its anticoagulant effects,³² and it binds both circulating and bound Xa. By limiting thrombin (IIa) formation, rivaroxaban prevents fibrin formation and, ultimately, clot formation. Indirectly through its effects on thrombin, rivaroxaban inhibits thrombin-induced platelet aggregation.³²

Antiplatelet and anticoagulant drugs are used to prevent thrombus propagation or recurrence in cats with aortic thromboembolism (ATE) associated with underlying hypertrophic cardiomyopathy (HCM). The preferred antithrombotic therapy for cats at risk of ATE is unclear, but options include clopidogrel (antiplatelet), low molecular weight heparin (anticoagulant) or rivaroxaban (anticoagulant), with dual therapy (antiplatelet and anticoagulant) sometimes administered.¹⁵⁶ Emerging studies support the use of rivaroxaban with or without clopidogrel to prevent thrombus formation in cats at risk of ATE and report few adverse effects.^{157, 159} In healthy cats treated with clopidogrel and rivaroxaban, platelet responsiveness to agonists, platelet activation and thrombin generation were reduced compared with treatment with either drug alone.¹⁵⁸ In a small prospective study comparing rivaroxaban with clopidogrel combined with enoxaparin in cats with HCM-associated ATE, rivaroxaban-treated cats showed no bleeding or thrombus recurrence.¹⁵⁷ In a multicenter prospective study evaluating recurrence of ATE with rivaroxaban or clopidogrel, the ATE recurrence rate of rivaroxaban (39%) was similar to clopidogrel (37%).¹⁵⁹ A small retrospective study in cats presenting for an ATE event or diagnosed with a cardiac thrombus reported a low rate of ATE recurrence (16.7%) when treated with both clopidogrel and rivaroxaban.¹⁶⁰

Although the formulation as a tablet may pose an administration challenge for cats, the available pharmacokinetics and pharmacodynamics of rivaroxaban in healthy cats indicate that it is well tolerated.^{32,157} Like in people, rivaroxaban peak plasma concentrations in cats occur approximately 2.5 h after oral dosing, with a dose-dependent increase in anti-Xa activity (Table 2).³² In people, rivaroxaban undergoes hepatic metabolism and is eliminated via feces and urine with both the active drug and its metabolites renally excreted.¹⁶¹ It is recommended to avoid prescribing rivaroxaban in cats with severe liver disease, and to use it with caution in cats with underlying kidney disease.³² Rivaroxaban is a p-glyco-protein substrate; coadministration with a p-glycoprotein inhibitor (eg, ciclosporin, azole antifungals, erythromycin) may increase plasma rivaroxaban concentrations and prolong its anticoagulant effects.^{32,33,162} In people, abrupt discontinuation of rivaroxaban can precipitate thrombotic events, so tapering is recommended before discontinuation.^{33,162}

Adverse effects reported in cats receiving rivaroxaban are summarized in Table 1.^{33,162} Spontaneous bleeding has not been reported in published studies involving healthy cats or cats with ATEs receiving rivaroxaban alone.^{32,157,158} However, when rivaroxaban was combined with clopidogrel in cats at risk for ATEs, clinical bleeding was observed in 15%, and included cases of epistaxis, hematuria and gastrointestinal bleeding.¹⁵⁸

Despite limited studies, rivaroxaban shows promise as an anticoagulant option for cats, either as a monotherapy or in combination with clopidogrel. The interest in rivaroxaban in cats at risk for ATE continues to grow, as it provides an oral anticoagulant option that avoids the need for daily injections of low molecular weight heparin and potentially reduces the need for therapeutic drug monitoring.

Capmorelin

Capmorelin is a growth hormone secretagogue (GHS) that stimulates appetite by acting as a selective agonist of the ghrelin receptor (or GHS-receptor 1a) and promoting growth hormone secretion via the hypothalamus and pituitary (Figure 8).¹⁶³ Capmorelin (Elura; Elanco Animal Health) is a veterinary formulated ghrelin receptor agonist for managing weight loss in cats with CKD and is used off-label for appetite stimulation and weight loss management in cats without CKD (Table 2).^{34,164,167} Its formulation as a liquid suspension and small volume given allow for easy administration. Capmorelin-treated healthy laboratory cats gained weight, increased food intake and had elevated concentrations of insulin-like growth factor-1 (IGF-1) compared with placebo-treated cats.^{165,166}

Figure 8

A simplified schematic representation of the role of ghrelin (a growth hormone secretagogue [GHS]) in appetite regulation and the mechanism of action of growth hormone secretagogues (eg, capmorelin). During fasting, the stomach produces ghrelin, which acts via the hypothalamus and peripherally to stimulate appetite. Ghrelin binds the ghrelin receptor (GR or GHS-receptor) on the hypothalamus and pituitary to produce growth hormone.¹⁶³ Indirectly, the hypothalamus triggers growth hormone production through growth hormone-releasing hormone (GHRH) binding the GHRH receptor in the pituitary. Growth hormone prompts the liver to secrete insulin-like growth factor-1 (IGF-1), leading to an increase in muscle mass.¹⁶³ Ghrelin, at least transiently, decreases glucose-stimulated insulin secretion via the GR on the delta cells of the pancreas, resulting in variable increases in blood glucose concentrations.¹⁶⁸ Capmorelin is a GR agonist, selectively targeting the GHS-receptor 1a to augment growth hormone secretion and increase IGF-1 concentrations, collectively improving appetite and promoting weight gain.^{164,165}

Capmorelin is absorbed in the proximal gastrointestinal tract, metabolized by hepatic enzymes and excreted primarily in the feces within 72 h, with minor urinary excretion in dogs.¹⁶⁴ It has not been studied in cats with hepatic disease. In fasted, healthy cats, peak concentrations occur 0.25–1 h after administration, with a half-life of 1.1 h (Table 2).³⁴ When given with food, the time to peak concentrations is delayed and reduced by approximately 50%.³⁴ Reported capmorelin adverse effects are listed in Table 1.^{34,166,168}

Capmorelin offers an effective oral solution for short-term appetite stimulation in anorexic cats. However, capmorelin affects pancreatic delta cells in cats, reducing glucose-stimulated insulin secretion and leading to variable increases in glucose concentrations;¹⁶⁸ in healthy cats, capmorelin causes transient increases in serum glucose and IGF-1 concentrations.^{34,167,168} Capmorelin is therefore contraindicated in cats with acromegaly and should be used cautiously in cats with current or historical diabetes mellitus, due to its impact on glucose metabolism and IGF-1 concentrations.^{34,167,168} Capmorelin should also be avoided in cats at risk for hypotension due to the potential effects low molecular ghrelin receptor agonists have on vascular receptors.¹⁶⁹

Rapamycin

Rapamycin (sirolimus) is a macrolide that appears to have beneficial effects on the progression of feline HCM, and may also offer potential benefits for other chronic conditions such as CKD and neoplasia. Rapamycin inhibits the mechanistic target of rapamycin (mTOR), a serine/threonine protein kinase present in two multiprotein complexes: mTOR complex 1 (mTORC1) and mTOR

complex 2 (mTORC2) (Figure 9). It is generally accepted that the benefits of rapamycin are primarily attributed to its inhibition of mTORC1, while many adverse effects arise from mTORC2 inhibition.^{170,171}

Figure 9

Highly simplified diagram representing the differences between the mechanistic target of rapamycin complex 1 (mTORC1) and mTORC2. Understanding of the intricate physiology associated with mTOR continues to evolve and is described in detail elsewhere.^{170,176} mTORC1 is largely a positive regulator of several anabolic processes, including protein and lipid synthesis, but it is also a negative regulator of autophagy. Less is known about the activity of mTORC2, but it is involved in lipid and glucose metabolism as well as cell survival, proliferation and apoptosis. Along with differences in the functions of these two complexes, there are differences in their sensitivities to rapamycin. Rapamycin acutely inhibits mTORC1, whereas mTORC2 is less sensitive, with inhibition occurring after chronic exposure. The goal of intermittent dosing of rapamycin is to primarily inhibit mTORC1 while minimizing inhibition of mTORC2

The mTOR signaling pathway plays a key role in the multifactorial process of aging. Rapamycin's antiaging effects are believed to result from its modulation of protein homeostasis, endoplasmic reticulum stress, mitochondrial respiration and autophagy, which may offer potential benefits for various chronic diseases (Table 1).¹⁷² A clinical trial is currently underway to investigate its effects on CKD in cats.

Few studies have evaluated rapamycin's effects as an immunosuppressive or anticancer agent in feline medicine. Rapamycin inhibits feline lymphocyte proliferation in vitro; however, its immunosuppressive effects have not yet been studied in vivo.¹⁷³ mTOR and its activated form, phosphor-mTOR, are expressed more frequently in feline triple negative (ie, absence of estrogen and progesterone receptor expression and human epidermal growth factor 2 [HER2] overexpression) as well as HER2-negative feline mammary carcinomas.¹⁷⁴ Additionally, the mTOR pathway is frequently activated with feline squamous cell carcinomas.¹⁷⁵ Given that mTORC1 activity promotes protein synthesis, cell growth and proliferation, this pathway may represent a therapeutic target for a subset of feline mammary carcinomas, squamous cell carcinomas and potentially other neoplasms.

Rapamycin appears to have beneficial effects on the progression of feline HCM. Sirolimus (TriviumVet; rapamycin) delayed-release tablets (felycin-CA1) have recently been granted conditional approval for the management of ventricular hypertrophy in cats with subclinical HCM.³⁵ The diagnosis of subclinical HCM can be made in a cat with a left ventricular wall thickness at end diastole of ≥ 6 mm that does not have systemic hypertension, other causes of compensatory myocardial hypertrophy, current or past clinical signs of congestive heart failure, ATE or severe left ventricle outflow tract obstruction.³⁵ A multiomics study in a colony of research cats with HCM showed that delayed-release rapamycin, given orally once weekly at a low dose (0.15 mg/kg) or high dose (0.3 mg/kg) for 60 days, resulted in dose-dependent suppression of myocardial hypertrophy and increased autophagic activity, and had anti-inflammatory and potential anticoagulant effects.³⁶ The exact mechanism of rapamycin's effects on cardiac remodeling is unknown, but it has an inhibitory effect on left ventricular hypertrophy in cats. In a prospective, randomized, double-masked, placebo-controlled study, delayed-release rapamycin was administered orally once weekly at a low dose (0.25–0.38 mg/kg) or high dose (0.52–0.73 mg/kg) over 180 days to cats with subclinical, non-obstructive HCM.³⁷ Cats receiving the low-dose rapamycin had a significantly lower maximum left ventricular wall thickness compared with those receiving placebo by the end of the study, with no differences in adverse events noted among treatment groups. Further research is needed to determine if rapamycin delays or prevents progression to congestive heart failure and improves survival.

Table 1 summarizes the adverse effects associated with rapamycin.^{170,171} The dosing strategies used in two feline HCM studies seem to minimize adverse effects;^{36,37} however, of the 30 cats treated with rapamycin in one of these studies, one developed hyperglycemia and one developed DKA, and they died after 1 month and 5 months of therapy, respectively.³⁷ The onset of diabetes mellitus in cats treated with rapamycin is of concern. Additionally, while its formulation as a tablet could pose administration challenges for cats, it only needs to be administered once a week. As the potential uses of rapamycin in cats become better understood, its profile and frequency of adverse effects should also become clearer.

Summary

As the landscape of feline medicine continues to evolve, the development of new drugs must be accompanied by proper implementation in clinical practice. It is essential for feline practitioners to stay informed about emerging therapies. Continued use of these pharmaceutical agents will provide a better understanding of their potential long-term benefits and adverse effects. As demonstrated in this review, new insights into established therapies are often being reported. For convenience, the 12 feline therapeutics discussed in this manuscript are summarized in Table 1, including the recommended dosages along with clinical indications and adverse effects. In addition, Table 2 summarizes the clinically relevant available pharmacokinetic data for these feline therapeutics.

Key Points

- ❖ Recent advancements in feline medicine are transforming disease management, making it essential for veterinarians to understand both the clinical application and regulatory considerations of novel therapies.

- ❖ Maintaining current knowledge of veterinary pharmaceuticals is critical for ensuring safe and effective treatment, as ongoing research continues to refine drug indications, dosages and safety profiles.
- ❖ Close clinical monitoring is vital when implementing new therapies, allowing for early detection of adverse effects and ensuring optimal therapeutic outcomes.

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